

Task 6 : Realtime Conversation with Voice Translation

Introduction

Realtime voice translation is an advanced application of machine learning that combines speech recognition, natural language processing, and sequence-to-sequence modeling to enable communication between speakers of different languages. This project focuses on enabling a real-time conversational environment between an English-speaking user and a Spanish-speaking user using custom-built machine learning models.

The system is designed to accept spoken input from users, convert speech into text, translate the text into the target language, and finally read the translated output aloud. The emphasis of this project is not on perfect translation accuracy, but on demonstrating technical depth, system integration, and end-to-end effort.

Background

Language barriers remain a major challenge in global communication. While many commercial translation tools exist, building such systems from scratch provides valuable insight into how modern machine learning models operate internally. This project draws inspiration from encoder-decoder architectures commonly used in neural machine translation systems.

To simulate a real-world bilingual conversation, two separate translation pipelines were implemented: Spanish-to-English and English-to-Spanish. Each pipeline uses independently trained sequence-to-sequence models, allowing bidirectional communication between users speaking different languages.

Learning Objectives

The learning objective of this project was to understand how speech-based machine translation systems are built from the ground up. This includes learning how to preprocess multilingual datasets, train neural networks for translation, and integrate trained models into a real-time application.

Another important objective was to gain hands-on experience with attention mechanisms, voice input handling, and text-to-speech synthesis. By combining multiple AI components into one system, this project strengthens practical understanding of applied machine learning.

Activities and Tasks

The project began with dataset preparation, including text normalization, tokenization, and the addition of special start and end tokens for sequence modeling. Separate datasets and tokenizers were prepared for English-to-Spanish and Spanish-to-English translation tasks.

Subsequently, two encoder-decoder LSTM models were trained. One model incorporated an attention mechanism to enhance contextual understanding. After training, both models were integrated into a single application that handles speech recognition, translation, and audio output.

ENG TO ESP :

Data with start and end tokens added:

	en	es
0	it's only three days till christmas.	<start> solo faltan tres días para navidad. <end>
1	harvard was founded in 1636.	<start> harvard se fundó en 1636. <end>
2	why do you waste most of your time on tatoeba?	<start> ¿por qué desperdicias la mayoría del t...
3	it is magnificent indeed.	<start> eso sí que es grandioso. <end>
4	she traveled all over the world.	<start> ella viajó por todo el mundo. <end>
...	...	...
14995	do you like chocolate milk?	<start> ¿te gusta la leche con chocolate? <end>
14996	tom set a trap.	<start> tom me puso una trampa. <end>
14997	i'll never eat it again.	<start> no lo volveré a comer. <end>
14998	i sweat every day.	<start> sudo todos los días. <end>
14999	my wife died of cancer.	<start> mi esposa murió de cáncer. <end>

15000 rows × 2 columns

Model :

Model: "model"			
Layer (type)	Output Shape	Param #	Connected to
=====			
input_1 (InputLayer)	[(None, 19)]	0	[]
input_2 (InputLayer)	[(None, 20)]	0	[]
embedding (Embedding)	(None, 19, 256)	3064320	['input_1[0][0]']
embedding_1 (Embedding)	(None, 20, 256)	4219392	['input_2[0][0]']
lstm (LSTM)	[(None, 19, 256), (None, 256), (None, 256)]	525312	['embedding[0][0]']
lstm_1 (LSTM)	[(None, 20, 256), (None, 256), (None, 256)]	525312	['embedding_1[0][0]', 'lstm[0][1]', 'lstm[0][2]']
bahdanau_attention (BahdanauAttention)	(None, 20, 256)	131841	['lstm[0][0]', 'lstm_1[0][0]']
concatenate (Concatenate)	(None, 20, 512)	0	['lstm_1[0][0]', 'bahdanau_attention[0][0]']
...			
Total params: 16,921,443			
Trainable params: 16,921,443			
Non-trainable params: 0			

Testing :

```
SRC: Tom put his phone in his pocket.  
PRED: tom puso su teléfono en su bolsillo.  
ACTUAL : Tom puso su teléfono en su bolsillo.  
  
SRC: He is slowly catching up.  
PRED: el está lentamente alcanzándole.  
ACTUAL : El está lentamente alcanzándole.  
  
SRC: Don't you get bored when you're alone?  
PRED: ¿no te aburres cuando estás sola?  
ACTUAL : ¿No te aburres cuando estás sola?  
  
SRC: He always does the opposite of what I tell him to do.  
PRED: él siempre hace lo opuesto de lo que le digo.  
ACTUAL : Él siempre hace lo opuesto de lo que le digo.  
  
SRC: You're cool.  
PRED: sos un langa.  
ACTUAL : Sos un langa.
```

ESP TO ENG :

Data :

	en	es
0	It's only three days till Christmas.	Solo faltan tres días para Navidad.
1	Harvard was founded in 1636.	Harvard se fundó en 1636.
2	Why do you waste most of your time on Tatoeba?	¿Por qué desperdicias la mayoría del tiempo en...
3	It is magnificent indeed.	Eso sí que es grandioso.
4	She traveled all over the world.	Ella viajó por todo el mundo.
5	Tom didn't mean to hurt you.	Tom no pensaba hacerte daño.
6	Tom asked for a loan from Mary.	Tom le pidió un préstamo a Mary.
7	According to the long-term forecast, a mild wi...	Según el pronóstico del tiempo a largo plazo, ...
8	That sounds interesting. What did you tell her?	Eso suena interesante. ¿Qué le dijiste?
9	Do you have any?	¿Tienes?
10	Women are not permitted to drive cars in Saudi...	A las mujeres no se les permite manejar en Ara...
11	I don't even know who I am anymore.	Ya no me conozco.
12	This box is light enough to carry.	Esta caja es lo suficientemente ligera para po...
13	He was sick yesterday.	Ayer estaba enfermo.
14	Yolks are yellow.	La yema es amarilla.
15	I think it's a trap.	Creo que es una trampa.
16	You should eat nutritious food.	Debes comer alimentos nutritivos.

Model :

```
Model: "model"
```

Layer (type)	Output Shape	Param #	Connected to
input_1 (InputLayer)	[(None, None)]	0	[]
input_2 (InputLayer)	[(None, None)]	0	[]
embedding_2 (Embedding)	(None, None, 256)	3025664	['input_1[0][0]']
embedding_3 (Embedding)	(None, None, 256)	2089984	['input_2[0][0]']
lstm (LSTM)	[(None, 256), (None, 256), (None, 256)]	525312	['embedding_2[0][0]']
lstm_1 (LSTM)	[(None, None, 256), (None, 256), (None, 256)]	525312	['embedding_3[0][0]', 'lstm[0][1]', 'lstm[0][2]']
dense (Dense)	(None, None, 8164)	2098148	['lstm_1[0][0]']

```
=====
Total params: 8,264,420
Trainable params: 8,264,420
Non-trainable params: 0
=====
```

Testing:

```
--- Testing Dataset Samples ---

Spanish Input : La decisión depende de ti.
Predicted Eng  : it's up to us to him
Actual Eng     : It's up to you to decide.

Spanish Input : Tú me pides que seamos solo amigos, y a mí no me interesa ser tu amigo.
Predicted Eng  : you may miss your favorite sentence to be able to do but i'm asking
Actual Eng     : You're asking for us to just be friends, but I'm not interested in being your friend.

Spanish Input : Tatoeba no es un diccionario.
Predicted Eng  : good is a liar
Actual Eng     : Tatoeba isn't a dictionary.

Spanish Input : Tengo mucho que aprender de ti.
Predicted Eng  : i have a lot to learn of people
Actual Eng     : I have a lot to learn from you.

Spanish Input : No puedo aceptar eso.
Predicted Eng  : i can't accept
Actual Eng     : I can't accept that.
```

Skills and Competencies

This project strengthened skills in deep learning, particularly in sequence-to-sequence modeling using LSTMs. Implementing attention mechanisms and handling variable-length sequences improved understanding of advanced neural network concepts.

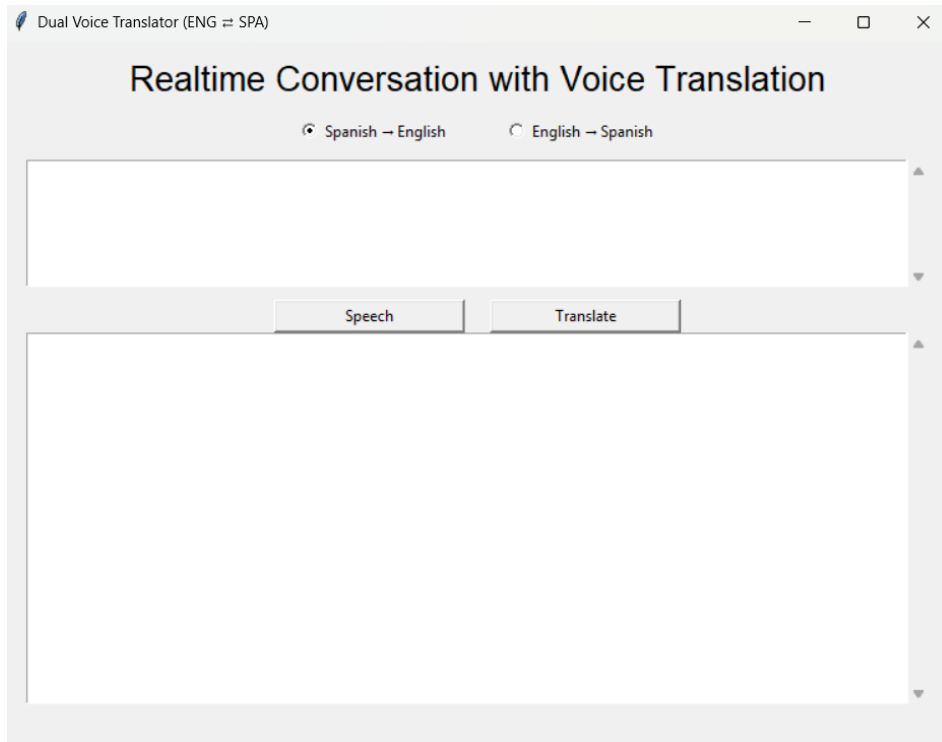
Additionally, the project enhanced practical skills in Python programming, TensorFlow, speech processing, and GUI development. Integrating multiple libraries such as Speech Recognition, gTTS, pyttsx3, and Tkinter demonstrated strong system-level thinking.

Feedback and Evidence

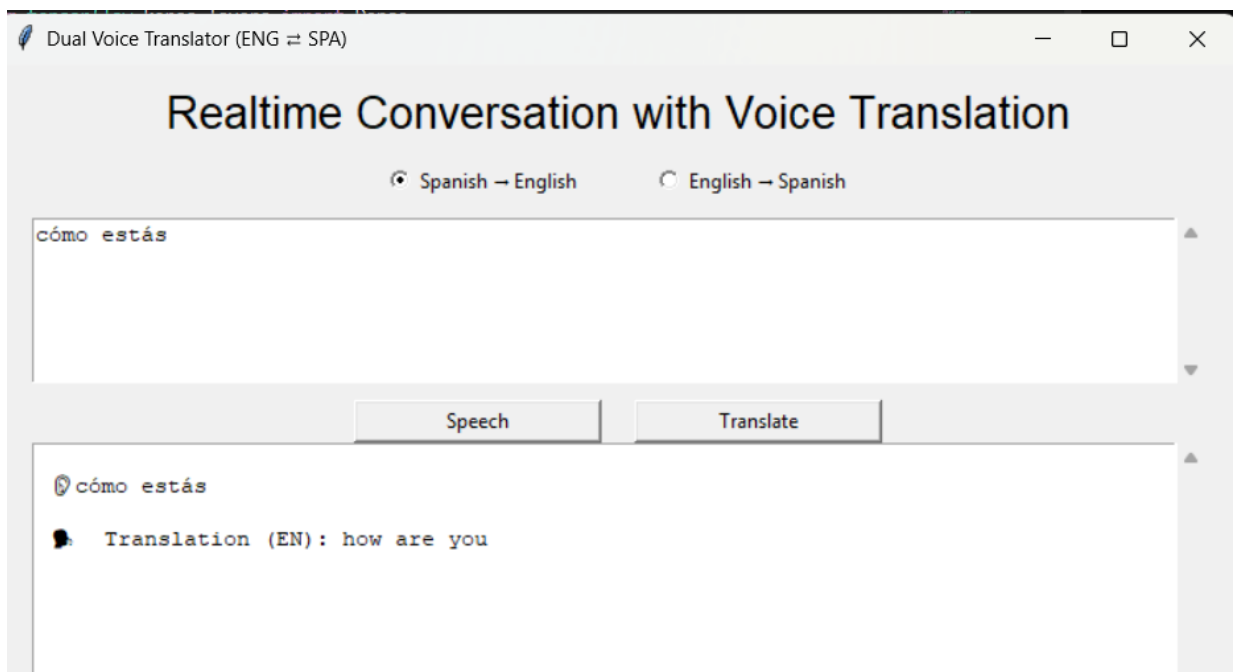
Model performance was evaluated through sample sentence testing rather than strict accuracy metrics, as per the project guidelines. The translations demonstrated reasonable word alignment and contextual relevance, confirming correct implementation of the translation pipeline.

Audio input and output functionality provided clear evidence of real-time interaction. The system successfully recognized spoken language, translated it, and generated audible output, validating the end-to-end workflow.

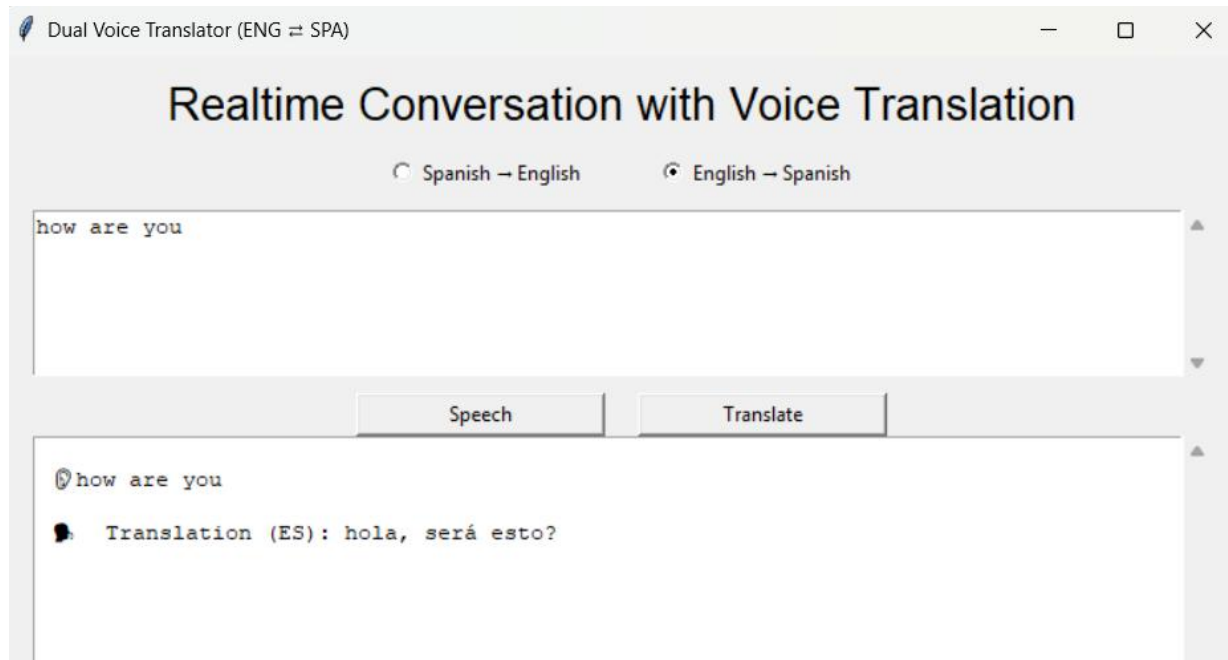
GUI :



GUI Working : **SPA to ENG:**



ENG TO SPA :



Challenges and Solutions

One major challenge was managing two separate translation models with different tokenizers and configurations. This was addressed by carefully organizing model files, tokenizer objects, and configuration parameters.

Another challenge involved real-time speech processing and audio playback synchronization. Threading and temporary file handling were used to ensure smooth playback without freezing the application interface.

Outcomes and Impact

The final system successfully enabled bilingual voice-based interaction between English and Spanish users. Despite limited dataset size and modest accuracy, the system demonstrated functional real-time translation.

This project highlights the practical feasibility of building conversational AI systems from scratch and serves as a strong foundation for future enhancements such as improved datasets, transformer models, or multilingual expansion.

Conclusion

This project demonstrates a complete real-time voice translation system using custom machine learning models. By combining speech recognition, neural translation, and text-to-speech synthesis, a functional bilingual communication tool was successfully developed.

More importantly, the project reflects significant effort, learning, and system integration. While accuracy was not the primary goal, the successful implementation of complex components showcases strong applied machine learning capability.